



STA 2101: Statistics & Probability

Lecture 02: Sampling Frequency Chart

Atanu Shuvam Roy



Lecturer,
Dept. of CSE, ULAB, Dhaka



Class Code
Sec 1 - es2uztok
Sec 5 - ayvzsvgm

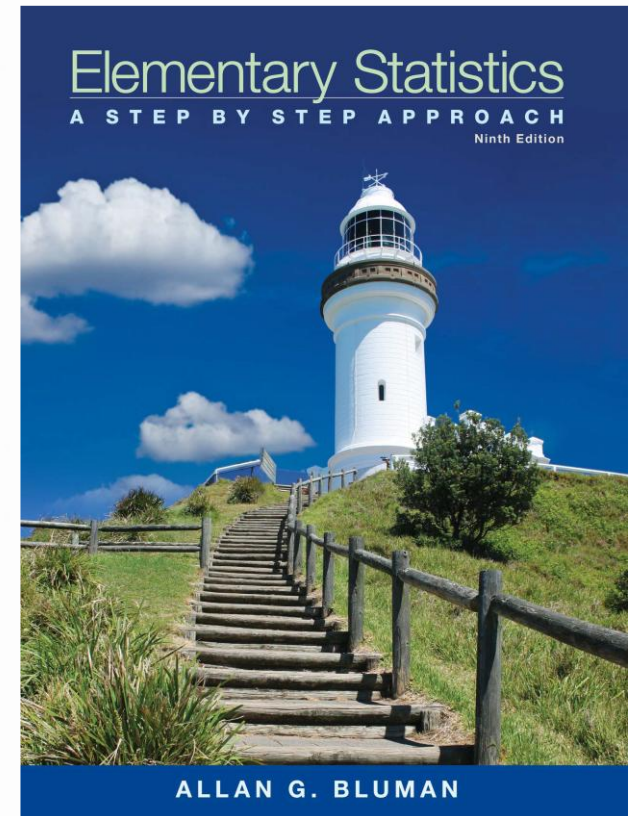


EMAIL:
atanu.shuvam@ulab.edu.bd

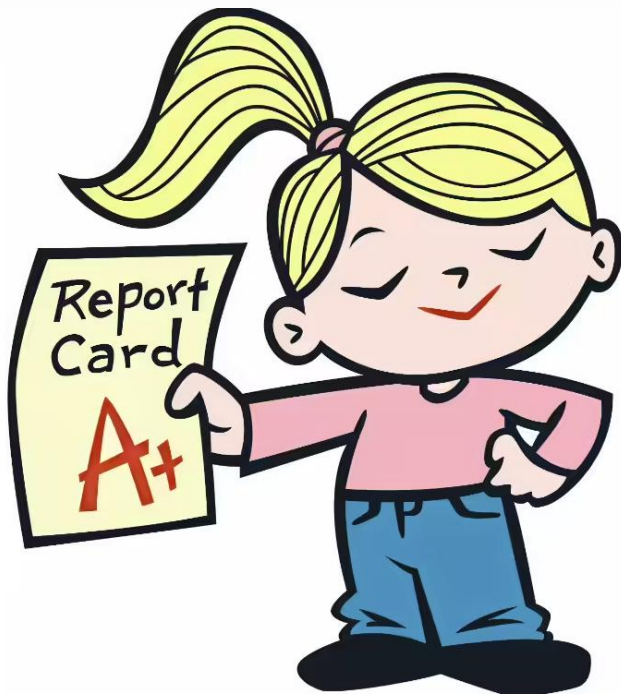
Spring 2026

Reference Book

- **Text Book(s):**
 - Elementary Statistics – Allan G. Bluman
 - Introduction to Probability and Statistics (14th Edition) By William Mendenhall, Robert J. Beaver, Barbara M. Beaver
- **Reference Material/Book(s):**
 - S Introduction to probability and statistics for engineers and scientists, Sheldon M Ross



Mark Distribution



Assessments	%
Mid Term Examination	25
Final Examination	40
Attendance and Class Participation	5
CT	5
Project	25
Total	100

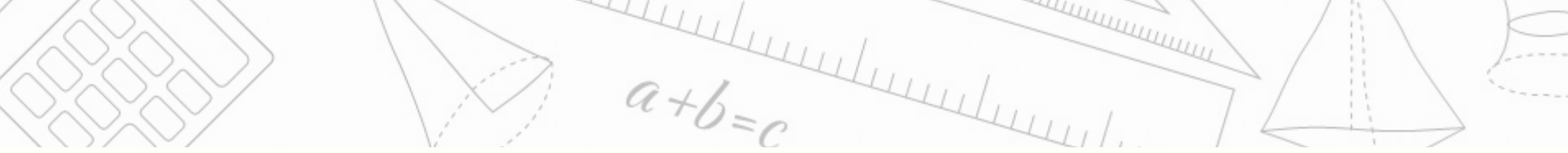
Today's Topics

- Sampling Methods

Sampling Methods

◆ Types of Sampling

1. **Probability Sampling** (each member has a known chance of selection)
 - **Simple Random Sampling**: equal chance for all
 - **Systematic Sampling**: selecting every kth element
 - **Stratified Sampling**: dividing into groups (strata) and sampling from each
 - **Cluster Sampling**: dividing into clusters, then randomly selecting clusters
2. **Non-Probability Sampling** (not every member has a chance of selection)
 - **Convenience Sampling**: based on ease of access
 - **Judgment/Purposive Sampling**: based on researcher's judgment
 - **Quota Sampling**: ensuring representation of certain groups
 - **Snowball Sampling**: participants recruit other participants



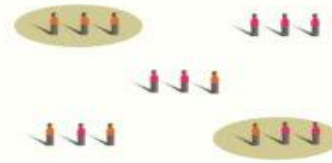
Probability Sample Methods



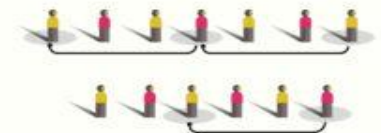
Simple Random Sampling



Stratified Random Sampling



Cluster Sampling

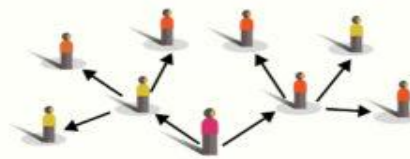


Systematic Sampling

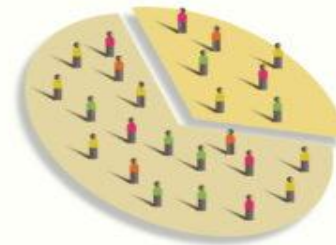
Non Probability Sample Methods



Convenience Sampling



Snowball Sampling



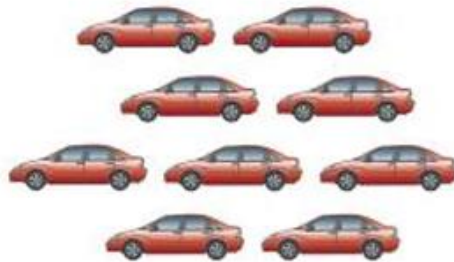
Quota Sampling



Purposive or
Judgmental Sampling

$$a+b=c$$

1. Random



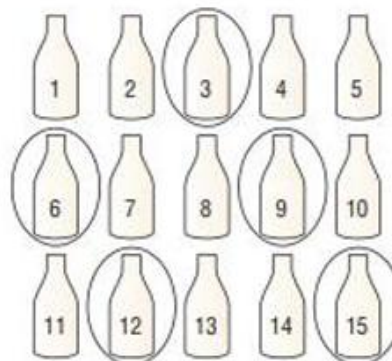
Population

Table D	Random Numbers			
10480	15011	01536	02011	81647
22368	46573	25595	85393	30995
24130	48360	22527	97265	76393
42167	93093	06243	61680	07856
37570	39975	81837	16656	06121
77921	06907	11008	42751	27750
99562	72905	56420	69994	98872
96301	91977	05463	07972	18876
89579	14342	63661	10281	17453
85475	36857	43342	53988	
28918	69578	88321		
63553	40961			



Sample

2. Systematic

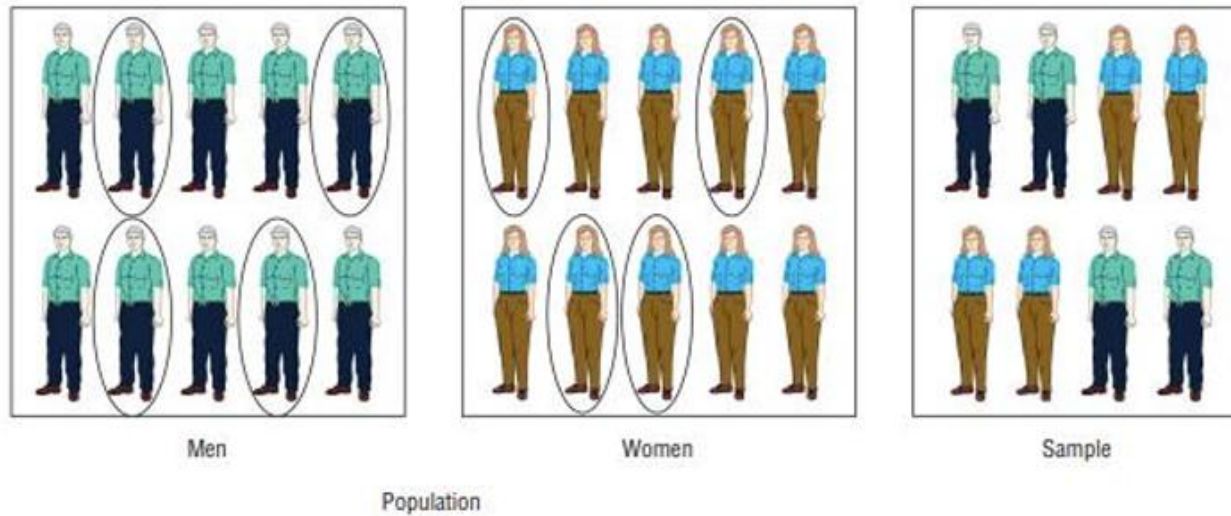


Population



Sample

3. Stratified



4. Cluster

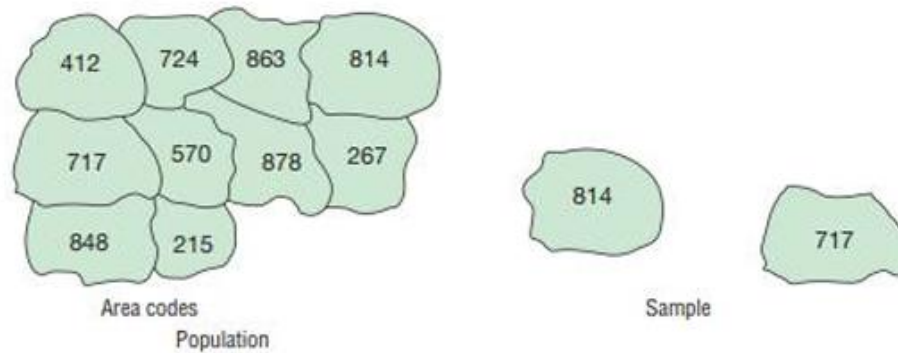


TABLE 1-4 Summary of Sampling Methods

Random	Subjects are selected by random numbers.
Systematic	Subjects are selected by using every k th number after the first subject is randomly selected from 1 through k .
Stratified	Subjects are selected by dividing up the population into subgroups (strata), and subjects are randomly selected within subgroups.
Cluster	Subjects are selected by using an intact subgroup that is representative of the population.

Example Practice

EXAMPLE 1-5 Sampling Methods

State which sampling method was used.

- Out of 10 hospitals in a municipality, a researcher selects one and collects records for a 24-hour period on the types of emergencies that were treated there.
- A researcher divides a group of students according to gender, major field, and low, average, and high grade point average. Then she randomly selects six students from each group to answer questions in a survey.
- The subscribers to a magazine are numbered. Then a sample of these people is selected using random numbers.
- Every 10th bottle of Super-Duper Cola is selected, and the amount of liquid in the bottle is measured. The purpose is to see if the machines that fill the bottles are working properly.

SOLUTION

- Cluster
- Stratified
- Random
- Systematic



ULAB
UNIVERSITY OF LIBERAL ARTS
BANGLADESH

Thank You

atanu.Shuvam@ulab.edu.bd

University of Liberal Arts Bangladesh
Department of Computer Science & Engineering

